

Chapter 00-B: Zero-Prerequisite Global Infrastructure Primer

The Bridge: Moving from Code to the Cloud! In Tracks 1 and 2, you mastered writing clean, thread-safe Python code.

But what happens when **100 million people simultaneously** use your application across London, Tokyo, New York, and Sydney?

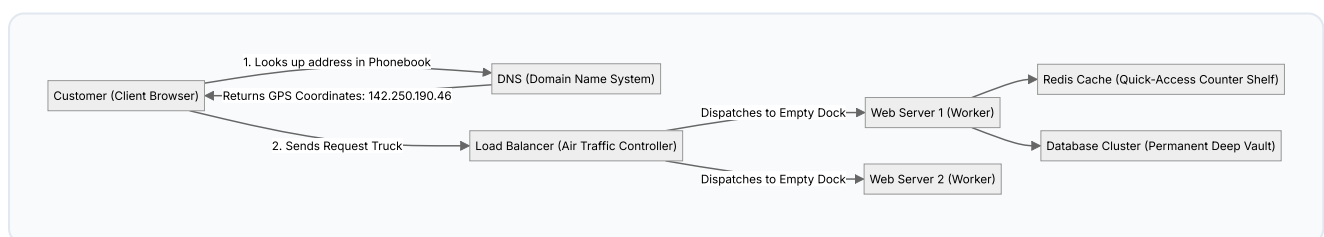
A single server—no matter how powerful—will melt under that traffic. Network cables under the Atlantic Ocean have physical speed-of-light latency limits (100ms). Hard drives fail. Entire AWS data centers lose power during hurricanes.

High-Level Design (HLD) is the architecture of planetary-scale distributed systems. It is the science of making 1,000 computers collaborate so seamlessly that to the user, they feel like one single, indestructible machine.

Before we design systems like YouTube, Uber, and WhatsApp, this primer will spoon-feed the foundational infrastructure vocabulary with zero jargon!

0. The Jargon Demystifier: The "Global Logistics & Postal Network"

Think of the internet not as mysterious cloud magic, but as a **worldwide shipping and delivery system**:



Distributed System Concept	Physical Postal Metaphor	Plain-English Definition
Client	The Customer at home	The mobile phone app or laptop web browser requesting information.
Server	A Postal Worker in a warehouse	A computer with high CPU/RAM and no monitor, sitting in a refrigerated data center.
IP Address	The Physical Postal Address	A unique digital street address (e.g. <code>192.168.1.1</code> or <code>142.250.190.46</code>).
DNS	The Universal Phonebook	Translates friendly human names (<code>google.com</code>) into computer IP addresses.
Latency	Delivery Transit Time	The number of milliseconds it takes for a data packet to travel round-trip.
Throughput (QPS)	Packages Processed per Second	Queries Per Second: how many requests your system can handle every second.
Load Balancer (LB)	The Air Traffic Controller	Stands in front of your servers; distributes incoming traffic evenly across healthy machines.
Reverse Proxy	The Security Guard at the Gate	Intercepts requests, checks credentials, compresses data, and shields backend servers from attackers.
CDN (Content Delivery Network)	Regional Neighborhood Warehouses	Caches static images/videos close to users (e.g. Tokyo edge server for Tokyo users).
Sharding	Splitting a giant 5,000-page dictionary	Dividing a massive database into smaller pieces (Volume A–M on Server 1, Volume N–Z on Server 2).
Replication	Photocopying documents in another city	Keeping identical copies of data on multiple servers in case one server catches fire.

1. Network Protocols: TCP vs. UDP (Spoon-Fed)

When two computers speak across the internet, they choose between two shipping modes:

1. TCP (Transmission Control Protocol) = "Registered Certified Mail"

- **How it works:** Sender and receiver perform a 3-way handshake (`SYN` → `SYN-ACK` → `ACK`). Every single packet is numbered. If Packet #3 is lost in the Atlantic Ocean, the receiver asks the

sender to re-transmit Packet #3 before reading Packet #4.

- **Guarantee: 100% Reliability & Strict Ordering.** Zero lost bytes.
- **Used for:** Bank transfers, text messages, database queries, and web page HTML.

2. UDP (User Datagram Protocol) = "Live Radio Broadcast"

- **How it works:** The sender fires packets into the network as fast as possible without waiting for acknowledgments. If 5 packets drop due to wifi interference, they are gone forever.
- **Guarantee: Maximum Speed & Minimum Latency**, but packets may be lost or arrive out of order.
- **Used for:** Live Zoom video calls, online multiplayer games, and VoIP (nobody cares if one video pixel dropped 50 milliseconds ago).

2. Load Balancing: Layer-4 vs. Layer-7 (Spoon-Fed)

Layer-7 Load Balancer (Application Level - Smart Routing)

Decrypts HTTPS & Reads URL Path & Headers:
If path == '/api/payments' → Send to Payment Cluster
If path == '/video/stream' → Send to Video Cluster

Layer-4 Load Balancer (Transport Level - Ultra Fast)

Inspects ONLY IP Address & Port (e.g. 192.168.1.5:443)
Does NOT read HTTP headers or URLs!
Blazingly fast (Millions of packets/sec)

- **Layer-4 (L4 - e.g. AWS NLB, IPVS):** Fast routing based strictly on raw TCP/UDP packet headers. It never decrypts HTTPS or looks inside the payload.
- **Layer-7 (L7 - e.g. NGINX, AWS ALB, Envoy):** Decrypts the request, inspects cookies, user headers, and URL routes, allowing smart routing, rate limiting, and SSL termination.

3. The 45-Minute HLD Interview Survival Script

When an interviewer asks: *"Design Netflix"* or *"Design Twitter"*, never jump into drawing boxes immediately! Follow this exact **5-minute opening script**:

Step 1: Scope & Functional Boundaries (Minutes 0 – 3)

"To ensure we build the right system, let's establish the core scope. Are we focusing on:

- 1. User video playback and streaming?*
- 2. Or the content creator video upload and transcoding pipeline?*
- 3. Should we design recommendations, or can we treat that as an external service?"*

Step 2: Back-of-the-Envelope Estimation (Minutes 3 – 5)

"Let's estimate the scale:

- Assume 100 Million Daily Active Users (DAU).
- Average write load: 10% of users upload 1 video/week $\rightarrow \sim 150$ QPS.
- Average read load: 5 videos watched per user/day $\rightarrow \frac{500 \times 10^6}{10^5} \approx 5,000$ QPS playback requests.
- *This is a **Read-Heavy System (ratio 33:1)**, which means aggressive multi-tier caching (Redis + CDN edge) is our highest architectural priority.*

4. Milestone Check: What You Just Mastered!

You now understand:

1. The global postal metaphor behind DNS, load balancers, and reverse proxies.
2. The exact mechanical difference between **TCP** (reliability) and **UDP** (speed).
3. Why companies use **Layer-4 for raw network routing** and **Layer-7 for smart application routing**.
4. The exact 5-minute opening script to dominate any High-Level Design interview.

Further Reading

- [System Design Primer](#)
 - [Cloudflare: what is a CDN](#)
 - [AWS: what is load balancing](#)
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Check Yourself

Answer in your head or on paper first, then open each answer.

▼ 1. What is the difference between latency and throughput?

Latency is time per request; throughput is requests per unit time. A system can have high throughput and high latency (batching).

▼ 2. What is the purpose of a load balancer?

Distribute requests across servers for capacity and availability, with health checks to remove bad instances.

▼ 3. Vertical versus horizontal scaling?

Vertical adds resources to one machine (limits, single point of failure); horizontal adds machines (needs stateless design or partitioning).

▼ 4. What does a CDN cache and why?

Static or cacheable content at edge locations near users to cut latency and origin load.